The zero endpoint is isolated from every positive normalized r-logSob index

A partial result for Open Problem (I) in arXiv:1108.1210

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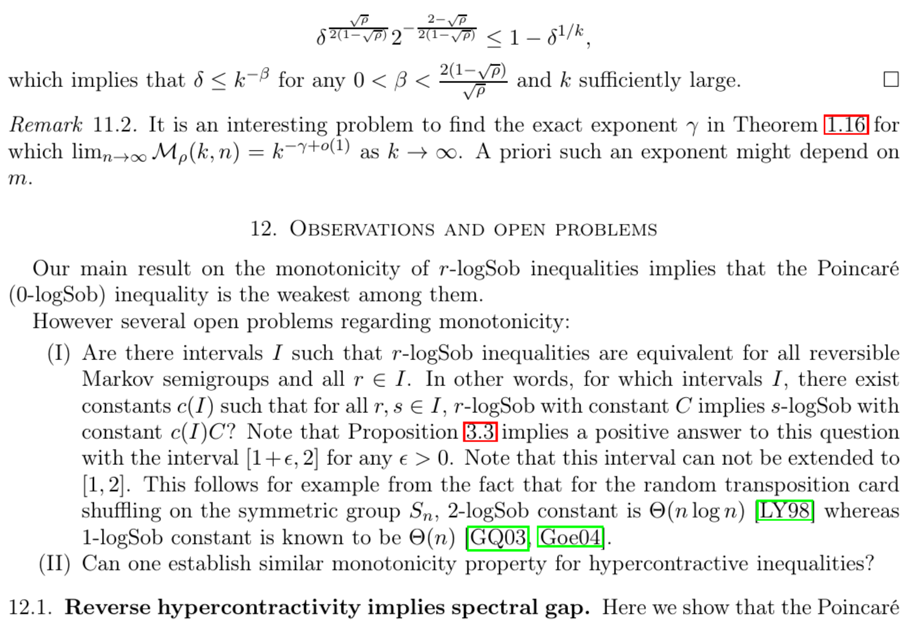
Source problem

Mossel, Oleszkiewicz, and Sen ask for which intervals I the normalized r-logSob inequalities are uniformly equivalent over all finite reversible Markov semigroups: a constant at one index should imply a constant at every other index with loss depending only on I [1, Open Problem (I), Section 12].



$$\delta^{\frac{\sqrt{\rho}}{2(1-\sqrt{\rho})}} 2^{-\frac{2-\sqrt{\rho}}{2(1-\sqrt{\rho})}} \leq 1 - \delta^{1/k},$$

which implies that $\delta \leq k^{-\beta}$ for any $0 < \beta < \frac{2(1-\sqrt{\rho})}{\sqrt{\rho}}$ and k sufficiently large. □

Remark 11.2. It is an interesting problem to find the exact exponent γ in Theorem 1.16 for which $\lim_{n\to\infty} \mathcal{M}_\rho(k,n) = k^{-\gamma+o(1)}$ as $k \to \infty$. A priori such an exponent might depend on m.

12. Observations and open problems

Our main result on the monotonicity of r-logSob inequalities implies that the Poincaré (0-logSob) inequality is the weakest among them.

However several open problems regarding monotonicity:

(I) Are there intervals I such that r-logSob inequalities are equivalent for all reversible Markov semigroups and all $r \in I$. In other words, for which intervals I, there exist constants $c(I)$ such that for all $r, s \in I$, r-logSob with constant C implies s-logSob with constant $c(I)C$? Note that Proposition 3.3 implies a positive answer to this question with the interval $[1+\epsilon, 2]$ for any $\epsilon > 0$. Note that this interval can not be extended to $[1, 2]$. This follows for example from the fact that for the random transposition card shuffling on the symmetric group S_n, 2-logSob constant is $\Theta(n \log n)$ [LY98] whereas 1-logSob constant is known to be $\Theta(n)$ [GQ03, Goe04].

(II) Can one establish similar monotonicity property for hypercontractive inequalities?

12.1. **Reverse hypercontractivity implies spectral gap.** Here we show that the Poincaré

Open Problem (I), official arXiv PDF page 27.

The source itself separates indices 0 and 1 using bounded-degree expanders. The result below fills the whole punctured interval between them on the zero-endpoint side.

Proof Intuition

An r-logSob inequality with fixed $r > 0$ gives reverse hypercontractivity after a time proportional to its constant. Testing the resulting two-function inequality on two single vertices forces a polynomial lower bound on every transition probability. In a bounded-degree graph, some vertices are logarithmically far apart, while a continuous-time walk can reach them before that many Poisson-distributed jumps only with a much smaller probability. Therefore the r-logSob constant must grow at least like $\log n$. Expanders keep the 0-logSob (Poincare) constant bounded, so zero cannot share an equivalence interval with any positive index.

Theorem. *Fix $0 < r < 1$ and an integer $d \geq 3$. There are constants $c_{r,d} > 0$ and $N_{r,d}$ such that the following holds. If G is a connected d-regular graph on $n \geq N_{r,d}$ vertices, P is its simple-random-walk kernel, and the semigroup generated by $L = I - P$ satisfies the normalized r-logSob inequality with constant C, then*

$$C \geq c_{r,d} \log n.$$

Proof. Put $s = r/2 \in (0, 1)$ and $s' = s/(s-1) < 0$. Proposition 6.1 of [1], applied with output exponent s' and input exponent s, gives

$$\|T_t h\|_{s'} \geq \|h\|_s, \qquad t = \kappa_r C, \qquad \kappa_r = -\tfrac{1}{2}\log(1-s) > 0.$$

Indeed the proposition's time is

$$\frac{C}{4} \log \frac{1-s'}{1-s} = -\frac{C}{2} \log(1-s).$$

Reverse Holder, followed by the displayed reverse-hypercontractive estimate, therefore yields for nonnegative f, g

$$\mathbb{E}_\pi[f T_t g] \geq \|f\|_s \, \|T_t g\|_{s'} \geq \|f\|_s \, \|g\|_s.$$

Take $f = \mathbf{1}_{\{u\}}$ and $g = \mathbf{1}_{\{v\}}$. Since the stationary measure is uniform,

$$\frac{1}{n} P_t(u, v) \geq n^{-2/s}, \qquad \text{hence} \qquad P_t(u, v) \geq n^{-A_r}, \quad A_r = \frac{4}{r} - 1. \tag{1}$$

A ball of radius R in a graph of maximum degree d has at most $1 + d((d-1)^R - 1)/(d-2)$ vertices. Consequently, for all sufficiently large n there are u, v at distance

$$D \geq a_d \log n$$

with a constant $a_d > 0$ depending only on d. The continuous-time walk makes a Poisson(t) number J of jumps. It cannot reach v from u in fewer than D jumps, and for $t < D$ the Chernoff bound gives

$$P_t(u, v) \leq \Pr\{J \geq D\} \leq \left(\frac{et}{D}\right)^D. \tag{2}$$

Choose $0 < \theta_{r,d} < e^{-1}$ so small that $a_d \log(e\theta_{r,d}) < -2A_r$. If $t \leq \theta_{r,d} D$, then (2) is at most n^{-2A_r}, contradicting (1). Thus

$$\kappa_r C = t > \theta_{r,d} D \geq \theta_{r,d} a_d \log n,$$

which proves the theorem with $c_{r,d} = \theta_{r,d} a_d / \kappa_r$. □

Corollary for Open Problem (I). *No interval $I \subset [0, 2]$ containing both 0 and a positive number has the equivalence property in Open Problem (I).*

Proof. For a fixed bounded-degree spectral expander family, the Poincare constants are uniformly bounded. Lemma 3.1 of [1] identifies this, up to its stated factor, with a uniform 0-logSob constant. For any fixed $r \in I \cap (0, 1)$ the theorem makes the optimal r-logSob constants diverge. If the positive point of I is at least one, interval convexity supplies a point in $(0, 1)$; alternatively the source already supplies the $r = 1$ separation. Hence uniform equivalence is impossible. □

Verification, limitations, and novelty

The exponent conversion, indicator normalization, bounded-degree diameter bound, and Poisson Chernoff comparison were independently rederived from the official source PDF. The algebra regression in `code/verify_expander_algebra.py` checks the chosen constants on a range of indices and degrees; it is not used as proof.

This settles only intervals that contain zero. It does not compare two fixed indices in $(0, 1)$ and does not classify the intervals bounded away from zero.

Cheap run indexes plus bounded exact-phrase, "r-logSob expander", positive-index, and reverse-hypercontractivity searches through 21 August 2026 found the source and later work on other reverse-hypercontractive settings, but no claimed resolution of this zero-versus-every-positive-index statement. Novelty remains subject to human review.

Human-review focus. Check the admissible exponent range in source Proposition 6.1, the factor κ_r, and the conversion from stationary expectation to the transition probability in (1). The remainder is the standard bounded-degree ball count and Poisson Chernoff bound.

References

[1] E. Mossel, K. Oleszkiewicz, and A. Sen, *On reverse hypercontractivity*, arXiv:1108.1210v2 (2018), especially Proposition 6.1 and Section 12.